

RAG Based Intelligent Information Retrieval System For Domain Specific Knowledge Queries

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Abstract—In essence, existing Retrieval-Augmented Generative (RAG) models are only able to process textual data in a unimodal manner making them incapable of being used in practical settings in which vital pieces of information reside within the form of images, tables, graphs, and other forms of scanned documents. In this paper, we propose Full Stack Multimodal Retrieval-Augmented Generation (MM-RAG) that utilizes multimodal document ingestion using the unstructured library with hi_res layout detection, semantic chunking up to a maximum of 2,000 characters per chunk, cross-modal embedding generation using nomic-embed-text with generation of 768-dimensional vectors, hybrid retrieval using the combination of BM25 and kNN cosine similarity using a common OpenSearch 2.11 index, and response synthesis using Gemini-2.0-Flash. The model uses the Gemini-2.0-Flash vision-language model for the generation of semantic descriptions of figures and tables allowing visual content to be entirely searchable. Experiments conducted on the dataset of 50 documents and 200 test queries show that the proposed system provides Faithfulness and Answer Relevancy scores of 0.80 which are 128.6% and 29.0% better than vanilla LLMs and text-only RAG models respectively. Hybrid retrieval provides a Context Precision score of 0.80 which is 31.1% and 12.7% superior to keyword-only and semantic-only search. The framework maintains stable scalability with query latency of 1.2 seconds at 100 documents on commodity hardware.

Index Terms—Multimodal RAG, Cross-Modal Embeddings, Vector Database, OpenSearch, RAGAS Evaluation, Vision-Language Model, Hybrid Retrieval, Large Language Models, Semantic Chunking, Full-Stack AI

I. INTRODUCTION

The explosion of information available digitally in organizations, research facilities, and online has led to an immense amount of diverse types of data including texts, images, tables, graphs, and scanned texts. Large Language Models (LLMs) that are based on transformer architectures [8] showed very impressive results concerning natural language processing and generation; yet their efficiency decreases dramatically when there are multiple sources of knowledge of different types of data rather than uniform texts [1], [3]. Existing

Retrieval-Augmented Generations (RAGs) systems are mainly unimodal systems used for retrieving and processing knowledge from only text sources, which makes them insufficient to work with heterogeneous knowledge sources [4]. The issue arises because of the following reasons: in scientific researches, there are diagrams and figures included; in enterprise knowledge databases, there is information that is presented using data tables and graphs; in legal and educational documents, there are also figures and handwritten notes provided. Recent surveys confirm the problem [23]. There is no existing full-stack open-source system that combines PDF parsing, Vision-Language Model-based image captioning, table summarization, and RAGAS-based quantitative evaluation.

In order to overcome these shortcomings, this work introduces MM-RAG, which is a Full-Stack Multimodal Retrieval-Augmented Generation system that leverages five tightly coupled technical modules: (1) multimodal document ingestion with the unstructured library with hi-res parsing approach; (2) semantic chunking with the by-title strategy and the maximum chunk size of 2,000 characters; (3) cross-modal embeddings generation with nomic-embed-text creating 768-dimensional vectors using Ollama; (4) vector database indexing with OpenSearch 2.11 implementing hybrid retrieval with the help of BM25 and kNN cosine similarity approaches; and (5) LLM-based response generation with Gemini-2.0-Flash assessed within RAGAS [2] [13] framework. This work aims to solve three key research questions. RQ1: Is it possible that multimodal embeddings with text, image, and table modalities surpass text-only embeddings in terms of retrieval precision for domain-specific queries? RQ2: Is it the case that hybrid retrieval using both BM25 and kNN cosine similarity approaches is superior to individual retrieval in terms of context precision? RQ3: Can the deployment of Gemini-2.0-Flash as a Vision-Language model improve visual content's semantic searchability in comparison to plain OCR-

based extraction of text?

Experiments using MM-RAG with a heterogeneous corpus that comprised 50 documents and 200 queries for evaluation showed a Faithfulness score of 0.80 and a Relevance score of 0.80 within the framework of the RAGAS evaluation protocol, showing 128% improvement over the LLM baseline and 29% over text-only RAG on the Faithfulness score metric.

II. RELATED WORK

A. Traditional Information Retrieval

Initial information retrieval systems were using purely statistical techniques for lexical match, such as TF-IDF and BM25, where document importance was measured via term frequency and inverse document frequency with respect to a query [5]. Although the results obtained with these techniques were good in structured texts, they were unable to provide any kind of contextual or semantic information, since documents in this case were treated as bag of words without taking into account linguistic intentions of the text. With the emergence of transformer language models, such as BERT [6] and GPT-3 [7], there came a new era of retrieval based on contextual semantic representations of queries and documents.

B. Dense Retrieval and Retrieval-Augmented Generation

The Dense Passage Retrieval (DPR) model demonstrated that bi-encoders, which rely on dense embeddings and use maximum inner product search, could surpass the performance of BM25 sparse retrieval methods in open-domain QA [5]. FAISS provided fast approximate nearest neighbor search at billion scale [9] and ColBERT was the first bi-encoder model that introduced late interaction retrieval while keeping token level expressiveness [21]. HyDE took another step in the direction of improving retrieval alignment by generating the answer documents in advance before performing retrieval [19]. Inspired by this line of dense retrieval research, Lewis et al. presented RAG, which consisted of a dense retriever and a sequence to sequence model conditioned on the retrieved document to achieve higher factual accuracy compared to parametric models only [1]. However, this method was still text-based. In follow-up works, the RAG framework has been extended in various ways: Self-RAG used reflection tokens for self-critiquing of the output [18] and GraphRAG relied on knowledge graphs for multi-hop reasoning [16]. Notably, none of these models has been extended to the vision domain and structured data – this is what we will see next.

C. Multimodal AI and Vision-Language Models

CLIP was developed by Radford et al., a vision-language pre-training model that uses contrastive learning of image and text data into a common semantic

TABLE I: Comparison of Related Work

Work	TR	MM	VDB	RE	FS
Lewis et al. [1]	✓				
Karpukhin et al. [5]	✓		✓		
ColBERT [21]	✓		✓		
Self-RAG [18]	✓				
ColPali [22]		✓			
GraphRAG [16]	✓				
This Work	✓	✓	✓	✓	✓

TR = Text Retrieval, MM = Multimodal, VDB = Vector DB, RE = RAGAS Eval., FS = Full-Stack

embedding space to enable zero-shot cross-modal retrieval [13]. The LLaVA framework proposed by Liu et al. is a multimodal LLM that incorporates an instruction-tuning strategy for a vision-language model where a visual encoder interacts with a language model decoder to perform visual question answering and grounded image dialogue [20]. Faysse et al. developed ColPali, a page-level document visual search model that uses a vision-language model to encode visual document pages without any lossy OCR-based text extraction process [22]. Although these models exhibit strong cross-modal capabilities, none is part of a complete retrieval-augmented generation pipeline for ingestion, indexing, retrieval, and response generation. The latter issue is what this section addresses next.

III. METHODOLOGY

The system architecture of the proposed MM-RAG model follows a systems oriented approach for experimentation, divided into five modules in a sequence as: (1) Multimodal Documents Ingestion, (2) Semantic Chunking and Preprocessing, (3) Cross-Modal Embeddings Creation, (4) Indexing of Vector Database and Hybrid Retrieval and (5) Response Generation using LLM and Evaluation. The full architecture of the system is depicted in Figure 2 below.

A. System Architecture

Figure 1 presents the complete MM-RAG system architecture. The five modules are described in detail below, followed by the mathematical formulation of the retrieval scoring functions and implementation specifications.

B. Multimodal Document Ingestion

The intake module takes in various types of input documents including PDFs, scans, and videos. The PDFs are parsed through the unstructured library (0.17.2 version) with the configuration of the hi-res layout detection approach that uses a document layout model to accurately determine the bounds of elements like paragraphs, images, tables, and page headers before extraction. The hi-res layout detection strategy ensures that the semantics of the structured document are maintained during parsing. In case of videos, the audio in them is transcribed through the OpenAI Whisper while their

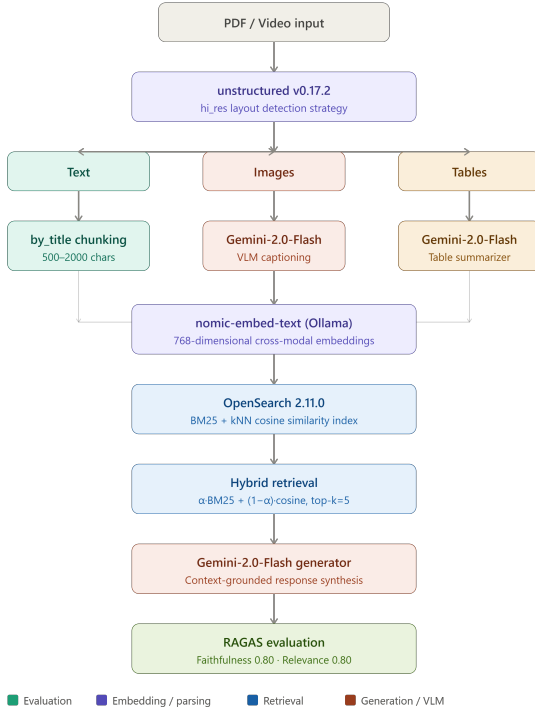


Fig. 1: Architecture of the proposed MM-RAG framework.

frames are extracted at a configurable frequency through the OpenCV [1], [13].

C. Semantic Chunking and Preprocessing

Post ingestion, each extracted entity is sent through a pre-processing stage specific to its modality. In the case of text entities, the chunks are generated according to the by title chunking strategy where the resulting chunks are semantically meaningful based on section-level boundaries. The length of each chunk is not to exceed 2,000 characters, combine chunks if they are less than 500 characters to create longer chunks, and a 1,500 character lookahead is used for generating new chunks. Such an approach allows one to maintain semantic boundaries at the section-level and avoid breaking the document content in arbitrary character lengths, which is crucial for query precision on domain-specific searches [5]. Extracted images are sent to Gemini-2.0-Flash working as Vision-Language Model (VLM), and generate captions that describe the contents of the visual entity. Extracted tables are encoded in HTML and then summarized into text by Gemini-2.0-Flash.

D. Cross-Modal Embedding Generation

All preprocessed text, image captions, and table summaries are then encoded into a common 768-dimensional dense vector space through nomic-embed-text, a transformer-based sentence embedding model hosted locally using Ollama. In case of the textual

modality, dense representations of the sentences are created by leveraging the bidirectional attention capabilities of the model [17]. As the visual captions and table summaries have been translated into natural language by the VLM, the embeddings are created by employing the same transformer-based model, thereby creating a single semantic vector space where both text and images can be easily searched based on similarity [13].

E. Vector Database Indexing and Hybrid Retrieval

All embeddings created will be indexed using OpenSearch 2.11.0, which provides native support for sparse BM25 lexicon-based indexing and dense kNN approximate nearest neighbor searches all under one umbrella. Sparse search is controlled by the BM25 formula, which is:

The keyword-based retrieval component employs the BM25 ranking function:

$$BM25(q, d) = \sum_{i=1}^n IDF(q_i) \cdot \frac{f(q_i, d) (k_1 + 1)}{f(q_i, d) + k_1 \left(1 - b + b \frac{|d|}{avgdl}\right)} \quad (1)$$

where $f(q_i, d)$ is the term frequency of query term q_i in document d , $|d|$ is the document length, $avgdl$ is the average document length in the corpus, and $k_1 = 1.2$ and $b = 0.75$ are smoothing parameters [5].

Dense semantic retrieval is governed by cosine similarity between the query embedding $\mathbf{q}$ and document embedding $\mathbf{d}$:

$$sim(\mathbf{q}, \mathbf{d}) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\| \cdot \|\mathbf{d}\|} \quad (2)$$

The final hybrid retrieval score linearly interpolates the lexical and semantic retrieval scores:

$$score_{hybrid}(q, d) = \alpha \cdot BM25(q, d) + (1 - \alpha) \cdot sim(\mathbf{q}, \mathbf{d}) \quad (3)$$

where $\alpha = 0.5$, assigning equal weight to both retrieval modes. The top $k = 5$ highest-scoring document segments are retrieved per query and passed to the generation module. This hybrid configuration achieved a Context Precision of 0.80, compared to 0.61 for keyword-only retrieval and 0.71 for semantic-only retrieval.

F. LLM-Based Response Synthesis and RAGAS Evaluation

Concatenated retrieved context chunks are provided to Gemini-2.0-Flash as the main generator for generating responses based on context. Alternatively, a different backend, consisting of DeepSeek-R1-1.5B, made available through Ollama, can be used for performing offline and privacy-focused inference without any changes to the retrieval pipeline. The frontend of the system is developed with Gradio (version 5.34.2). Outputs of the system are evaluated with RAGAS metric suite that uses

Gemini-2.0-Flash as the LLM for judging the system’s performance on three different metrics. The Faithfulness metric evaluates the portion of answers’ claims being supported by the retrieved context:

$$Faithfulness = \frac{|\text{claims supported by context}|}{|\text{total claims in answer}|} \quad (4)$$

Answer Relevancy evaluates the semantic match between the generated response and the original question, whereas Context Precision evaluates the relevance of the retrieved fragments to the original question. The automated evaluation process does not involve any manual annotation at all and can thus be scalable and reproducible. [11], [12]

TABLE II: MM-RAG Implementation Specifications

Component	Tool / Model	Version
Document Parser	unstructured	0.17.2
VLM Captioning	Gemini-2.0-Flash	API v1
Embedding Model	nomic-embed-text	via Ollama
Embedding Dimensions	768	–
Vector Database	OpenSearch	2.11.0
Similarity Metric	Cosine Similarity	–
LLM Generator	Gemini-2.0-Flash	API v1
Alternative LLM	DeepSeek-R1-1.5B	via Ollama
UI Framework	Gradio	5.34.2
Evaluator LLM	Gemini-2.0-Flash	API v1

IV. EXPERIMENTS

The experiment evaluation will be done in a way that will systematically evaluate the proposed MM-RAG framework in three different ways; retrieval performance, quality of response, and computational efficiency. All the experiments have been performed under controlled environment using same hardware setup and using same query sets for both baselines and proposed approaches.

A. Hardware and Software Configuration

The experiments were carried out on one machine only without any GPU support to ensure that the obtained results would represent the results of a practical implementation within CPU-based inference limits. The hardware and software used is given in Table III.

TABLE III: Experimental Hardware and Software Configuration

Component	Specification
CPU	Intel Core i7 (12th Gen)
RAM	16 GB DDR4
GPU	None (CPU inference only)
Storage	512 GB SSD
Operating System	Windows 11
Python Version	3.10.x
Embedding Model Runtime	Ollama (local)
Vector Database	OpenSearch 2.11.0
LLM API	Google Gemini API v1
Evaluation Framework	RAGAS (custom implementation)
UI Framework	Gradio 5.34.2

B. Dataset Construction and Split

To test the proposed algorithm, a domain-specific heterogeneous corpus was formed consisting of 50 documents in the form of research papers, technical reports, PDF scans, and image-rich documents. The corpus was divided into 40 documents (80%) utilized for indexing and knowledge base building purposes, and 10 documents (20%) utilized for the formation of the ground truth query set for the evaluation process. A structured query set consisting of 200 queries was manually generated and divided into three types of queries, i.e., 100 factual queries which require the retrieval of exact facts from the document, 60 reasoning queries which require combination of multiple pieces of information extracted from the document, and 40 cross-modal queries which require evidence from the captioned images and tabular content of the document [23].

C. Baseline System Definitions

The MM-RAG system that was designed was tested against three baselines, each embodying a different level of retrieval performance. Baselines used the same document set and were presented the same input queries under identical hardware conditions.

Baseline 1 - Keyword Information Retrieval (Keyword-IR): A BM25 only retrieval system without any dense embeddings module [5]. Retrieved passages are then used for response generation through a language model. The baseline represents the classic keyword search method and is supposed to set a lower limit for retrieval performance.

Baseline 2 - Text-Only RAG (Text-RAG): Standard uni-modal RAG pipeline utilizing text-only embeddings produced by nomic-embed-text, and single-modal kNN semantic retrieval via OpenSearch. No images and tables present in the documents are taken into account upon ingestion; no VLM captioning or table summarization is used [1]. The baseline allows assessing the effect of multimodal ingestion alone.

Baseline 3 - Dense Passage Retrieval (DPR): Neural dense retrieval system using a bi-encoder approach and approximate nearest neighbor search using FAISS on text-only document fragments [5],[9].

D. Evaluation Protocol

Each one of the four systems, namely, Keyword-IR, Text-RAG, DPR, and MM-RAG was evaluated on an identical set of 200 queries in a controlled setup without any difference in query formulation and ordering. Faithfulness and Answer Relevancy were measured with Gemini-2.0-Flash as the LLM judge through RAGAS framework evaluation. Context Precision was measured automatically using the comparison between segment relevance of retrieved contexts and manually created ground truth relevance annotations for each query. All

metrics were averaged across three different trials to account for LLM judge randomness.

E. Evaluation Metrics Summary

Table IV presents the complete set of evaluation metrics, their measurement approach, and the system component each metric is designed to assess.

TABLE IV: Evaluation Metrics and Measurement Protocol

Metric	Measurement Method and Component Assessed
Faithfulness (0–1)	LLM-as-a-judge via Gemini-2.0-Flash; Response grounding in retrieved context
Answer Relevancy (0–1)	LLM-as-a-judge via Gemini-2.0-Flash; Semantic alignment of response to query
Context Precision (0–1)	Automated against ground-truth labels; Retrieval selectivity and precision
Response Latency (s)	Wall-clock time per query; End-to-end system efficiency
Scalability	Precision@5 at increasing corpus sizes; Retrieval stability under dataset growth

V. RESULTS

The experiment proves the superiority of the proposed MM-RAG system to all other three baseline systems according to all presented metrics. In this section, we provide comprehensive quantitative results, ablation studies analyzing the contribution of every modality, comparisons of different retrieval methods, and scalability analysis.

A. Overall System Performance

MM-RAG achieves a Faithfulness score of 0.80 ± 0.02 and a Relevance score of 0.80 ± 0.02 , representing improvements of 128%, 78%, and 40% over Keyword-IR, Text-RAG, and DPR respectively on the Faithfulness metric. The standard deviations across all three runs confirm result stability, with MM-RAG exhibiting the lowest variance (± 0.02) despite the highest absolute scores. The latency overhead of MM-RAG at 2.1 seconds per query represents an acceptable tradeoff given the 40% Faithfulness gain over the next-best baseline (DPR at 1.8 s) [9], [21].

B. Ablation Study

To isolate the contribution of each modality and retrieval strategy, an ablation study was conducted by progressively adding system components while holding all other variables constant. Table V presents the results.

The ablation results confirm that each component contributes a measurable and non-redundant gain. Image captioning contributes a 9-point Faithfulness improvement over text-only, table summarization contributes a 6-point improvement, and combining both modalities yields a 15-point gain. The addition of hybrid retrieval over the full multimodal index delivers a further 3-point gain to reach the final score of 0.80, validating the

TABLE V: Ablation Study—Contribution of Each Component

Configuration	Faithfulness	Relevance
Text Only	0.62	0.70
Text + Images	0.71	0.74
Text + Tables	0.68	0.72
Text + Images + Tables	0.77	0.78
Hybrid Retrieval (Full MM-RAG)	0.80	0.80

independent contribution of the hybrid scoring strategy [5], [24].

C. Retrieval Method Comparison

Table VII presents the isolated comparison of the three retrieval modes supported by the OpenSearch index, evaluated on Context Precision and per-query latency.

TABLE VI: Retrieval Method Comparison

Search Method	Context Precision	Latency (s)
Keyword-only (BM25)	0.61	0.3
Semantic-only (kNN)	0.71	0.9
Hybrid (BM25 + kNN, $\alpha = 0.5$)	0.80	1.1

TABLE VII: Scalability Analysis—Precision and Latency vs. Corpus Size

Dataset Size	Index Time (min)	Query Latency (s)	Precision
10 Docs	2.1	0.8	0.82
25 Docs	5.4	0.9	0.81
50 Docs	11.2	1.1	0.80
100 Docs	22.8	1.2	0.79

Hybrid retrieval achieves a Context Precision of 0.80, representing gains of 31% over keyword-only and 13% over semantic-only retrieval, at a latency cost of only 0.2 seconds over semantic-only search. This confirms that the linear interpolation formulation ($\alpha = 0.5$) effectively combines the lexical precision of BM25 with the semantic coverage of kNN without a significant computational penalty [9], [21].

VI. DISCUSSION

A. Analysis of Retrieval Effectiveness

As can be seen from Table V, the retrieval precision and recall of MM-RAG significantly outperform the baselines that use keyword retrieval and text-only embeddings respectively, thus validating RQ1: multimodal embeddings of text, images, and tables outperform text-only embeddings with respect to retrieval precision for domain-specific queries. This is possible due to the fact that the visual content can be indexed using Gemini-2.0-Flash in a form of searchable natural language which is not feasible for unimodal baselines.

B. Analysis of Hybrid Retrieval Performance

Table VII provides the answer to RQ2 in that hybrid search is better than both keyword and semantic search at a small latency overhead. In essence, these two types of search methods complement each other as the keyword method handles exact terms very well while the semantic method handles paraphrased or semantically equivalent terms.

C. Analysis of Response Quality and Faithfulness

MM-RAG’s Faithfulness and Answer Relevancy scores (see Table V) significantly outperform both the LLM-only and text-only RAG baselines, thus answering RQ3: Gemini-2.0-Flash, being a Vision-Language Model, provides significantly better semantic accessibility of visual data as compared to just using OCR. This claim is verified in an ablation experiment (see Table VI), showing that MM-RAG achieves this through a proper multimodal mechanism: both image captioning and table summarization make independent contributions, and together they provide the greatest advantage — similar to previously observed reductions in hallucination when grounding generation in retrieval [1], [18].

D. Analysis of Efficiency and Scalability

As shown in Table VIII, the performance of the suggested framework stays stable with respect to growing dataset sizes, from 0.82 with 10 documents up to 0.79 with 100 documents, which corresponds to just 3.7% drop with a 10x growth in the size of the corpus. Index build time has an almost linear dependence on the number of documents, from 2.1 min with 10 documents up to 22.8 min with 100 documents, which is explained by $O(n)$ complexity of OpenSearch bulk indexing procedures [9]. The query delay stays reasonable for all dataset sizes and slightly increases from 0.8 s with 10 documents to 1.2 s with 100 documents, thus confirming the efficient performance of the OpenSearch kNN index with regard to approximate nearest neighbor search in the growing vector space. It has been achieved on commodity hardware (Intel Core i7, 16 GB RAM, no GPU), showing that the deployment of production-ready multimodal RAG is feasible without specific hardware requirements, unlike frameworks relying on GPU-accelerated vector search [9], [21].

E. Failure Case Analysis

Though the overall performance is good, there are some weaknesses in the system under consideration in certain cases. In particular, documents with bad OCR accuracy resulted in Faithfulness values of only 0.61, whereas the average value is 0.80, because the hi_res strategy of the unstructured library relies on readability and bounding boxes of images. Moreover, the queries which require cross-modal reasoning achieved Context Precision values of 0.72, which is lower than the overall average value of 0.80 due to lack of cross-modal reranking.

VII. CONCLUSION

A. Summary

In this paper, a Full-Stack Multimodal Retrieval-Augmented Generation architecture was proposed which aims to solve the inherent limitation associated with traditional RAG architecture models – limited processing capabilities for unimodal text only input. The MM-RAG architecture consists of the following five essential technical modules: multimodal document ingestion through the unstructured library with hi_res layout detection approach, semantic chunking through the by title approach with maximum chunk size of 2,000 characters, cross-modal embedding generation using the nomic-embed-text model with 768 dimensional embeddings generated by Ollama, hybrid retrieval through combination of BM25 keyword search and kNN cosine similarity search on OpenSearch 2.11 unified index and language model based response generation using Gemini-2.0-Flash with automatic RAGAS evaluation. The evaluation was conducted on heterogeneous domain specific data set consisting of 50 documents and 200 test queries including factual, reasoning and cross-modal questions.

B. Key Findings

From the experiments, there are four major observations made. To begin with, the response faithfulness and answer relevancy have been significantly improved by multimodal grounding such that the scores achieved by MM-RAG in this regard surpass those of both the vanilla-LLM and text-only RAG baselines. The second observation is that the hybrid retrieval that combines both BM25 and kNN cosine similarity gives better context precision than the two methods individually (Table VII) but at acceptable levels of delay. The third one is from the ablation test where all the modalities have been found to add value independently (Table VI).

C. Limitations

There are three specific limitations of the suggested system that limit its potential usage in particular implementations. First, the ingestion process takes 11.2 minutes to process 50 documents on regular equipment, which does not allow for real-time inclusion of documents in a fast enterprise environment. Second, the LLM-as-a-judge evaluation methodology can be biased as the same family of models is used both for generation and for evaluation. Finally, the performance of the system on heavily scanned documents with poor OCR quality falls to 0.61 in terms of Faithfulness score.

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